

Digital Queue & Appointment Management System

Problem-Solving Hackathon Project Draft

Core Domain: Digital queue, appointments, service counters, token management, waiting-time tracking, staff operations

PROJECT TYPE	CORE DOMAIN
Real-world service and appointment management application	Digital queue, appointments, service counters, token management, waiting-time tracking, staff operations
PLATFORM	FINAL OUTCOME
Web or Mobile Application	A deployed system that allows users to book appointments or receive digital queue tokens, track their waiting position, and helps organizations manage customers, counters, staff, and service flow efficiently.

1. Project Overview & Real-World Problem

The Digital Queue & Appointment Management System is designed for places such as:

- University offices
- Banks
- Clinics
- Government service centers
- Customer support offices
- Company HR departments
- Service centers

In many organizations, customers or visitors still stand physically in long queues without knowing how long they will have to wait.

Staff may also face problems such as:

- Too many people arriving at the same time
- No clear queue order
- Customers repeatedly asking when their turn will come
- Appointment and walk-in users being mixed together
- Missed appointments
- Overloaded service counters
- Poor record of service time
- Difficulty identifying peak hours

The system should allow users to either book an appointment or join a digital queue without standing physically in line.

The complete flow should be:

User Selects Service → Books Appointment or Gets Token → Queue Position Assigned → Waiting Time Estimated → Staff Calls User → Service Completed → Queue & Analytics Updated

2. Main Users & Their Roles

Main Users	Their Roles
Customer / Visitor	The user can: • Create an account • Select department or service • View available appointment slots • Book an appointment • Join a digital queue • Receive a token number • View current queue position • View estimated waiting time • Cancel or reschedule • Receive notifications • View previous appointments or visits
Service Staff	Staff can: • View waiting customers • Call the next token • Start service • Complete service • Skip unavailable customers • Recall a token • View customer appointment details • Update service status
Department Manager	The manager can: • Manage staff • Manage service counters • Create services • Set working hours • Define appointment duration • Set daily appointment limits • Monitor queue length • View staff workload • View waiting-time statistics
Administrator	The administrator can: • Manage departments • Manage users • Manage services • Manage permissions • View organization-wide activity • View reports and analytics

3. Appointment & Digital Queue Workflow

The system should support both:

Scheduled Appointments

and

Walk-In Digital Tokens

Appointment Example

A student wants to visit the university examination department.

The flow should be:

1. Student selects the department.
2. Student selects the required service.
3. Available dates are displayed.
4. Student chooses a time slot.
5. System checks availability.
6. Appointment is confirmed.
7. Student receives appointment number.
8. Reminder is sent before the appointment.
9. Student checks in on arrival.
10. Staff calls the student.
11. Service is completed.
12. Appointment is marked complete.

Possible statuses:

Booked → Confirmed → Checked In → Waiting → In Service → Completed

Other possible statuses:

- Cancelled
- Missed
- Rescheduled
- Delayed

Walk-In Queue Example

A visitor arrives without an appointment.

The user selects:

Service: Document Verification

The system generates:

Token: A-027

Current Token: A-021

People Ahead: 6

Estimated Waiting Time: 18 Minutes

The user can wait somewhere nearby instead of standing physically in line.

4. Smart Waiting Time & Queue Management

One intelligent feature can be Estimated Waiting Time.

Instead of only showing the token number, the system should estimate how long the user may need to wait.

Possible factors include:

- Number of people ahead
- Average service duration
- Number of active counters
- Current staff availability
- Appointment priority
- Current queue speed

Example:

People Ahead: 5
Average Service Time: 4 Minutes
Active Counters: 2

Estimated Waiting Time: 10 Minutes

The system should update this estimate as the queue changes.

Another important feature is Dynamic Queue Management.

If one counter closes, the system should redistribute waiting customers.

Example:

Counter 1 → Active
Counter 2 → Active
Counter 3 → Closed

The waiting-time estimate should automatically update.

5. Appointment Slot & Capacity Management

Each department should define:

- Working hours
- Number of service counters
- Appointment duration
- Maximum appointments per slot
- Break time
- Service availability

Example:

Department: Student Affairs

9:00–9:30 AM

Maximum Appointments: 6

Booked: 6

Status: Full

9:30–10:00 AM

Maximum Appointments: 6

Booked: 3

Status: Available

The system should prevent overbooking.

Different services may also require different time durations.

Example:

Document Collection → 5 Minutes

Certificate Verification → 10 Minutes

New Registration → 20 Minutes

This should be considered when assigning appointments.

6. Counter, Staff & Service Management

Organizations should be able to create multiple service counters.

Example:

Counter 1 → Document Verification
Counter 2 → Registration
Counter 3 → Fee Queries

Staff should be assigned to specific counters or services.

Possible counter statuses:

Available / Busy / Break / Closed

Possible staff information:

- Staff ID
- Name
- Department
- Assigned counter
- Service type
- Shift
- Current status

The system should also allow staff to temporarily pause a counter.

When this happens, the queue should adjust automatically.

7. Token Calling, Check-In & No-Show Handling

When a user's turn arrives, staff should call the token.

Example:

Token A-027 — Please proceed to Counter 3.

The user can receive:

- In-app notification
- Screen notification
- Email
- SMS if implemented

If the user does not appear, staff can:

- Recall token
- Skip token
- Mark as missed

Example:

A-027 → Called
A-027 → No Response
A-027 → Recalled
A-027 → Missed

For appointments, the system may define a check-in window.

Example:

Check in between 10 minutes before and 10 minutes after your appointment time.

If the user does not check in, the appointment can be marked as:

Missed

and the slot becomes available where appropriate.

8. Data Model & System Architecture

User Data	Service Data	Appointment Data	Token Data	Counter Data
Possible fields: • user_id • name • email • phone • role • account_status	Possible fields: • service_id • service_name • department_id • average_duration • active_status	Possible fields: • appointment_id • user_id • service_id • appointment_date • start_time • end_time • appointment_status • check_in_time	Possible fields: • token_id • token_number • user_id • service_id • queue_position • estimated_wait • token_status • created_at	Possible fields: • counter_id • department_id • assigned_staff • service_type • current_token • status

Recommended architecture:

Web / Mobile Application → Backend API → Appointment Manager → Queue & Token Manager → Database → Staff / Admin Dashboard

9. Search, Dashboard & Analytics

Users should be able to search by:

- Department
- Service
- Date
- Available appointment
- Appointment status

Management should have a dashboard showing:

- Total appointments today
- Walk-in visitors
- Current waiting customers
- Active counters
- Completed services
- Missed appointments
- Average waiting time
- Average service duration
- Busiest department
- Busiest service
- Peak visiting hours

Example:

Appointments Today: 84
 Walk-In Tokens: 126
 Currently Waiting: 17
 Active Counters: 6
 Average Waiting Time: 12 Minutes

Analytics may also include:

- Waiting time by department
- Queue length by hour
- Staff workload
- Service completion time
- Appointment cancellation rate
- No-show rate
- Daily and weekly visitor trends

10. Notifications, Rules & Error Handling

The system should send notifications when:

- Appointment is confirmed
- Appointment time is approaching
- Queue position is getting close
- User is called to a counter
- Appointment is rescheduled
- Service is delayed
- Appointment is cancelled

The application should also include:

- User authentication
- Role-based access
- Input validation
- Appointment history
- Queue history
- Staff activity logs
- Prevention of duplicate appointments
- Prevention of duplicate active tokens
- Error handling for failed bookings

Organizations should also be able to define rules such as:

- Maximum appointments per user per day
- Maximum queue tokens per user
- Appointment cancellation limit
- Late check-in handling
- Priority-service rules
- Department-specific booking rules

11. Final Project Vision & Additional AI Features

The final system should demonstrate:

User Selects Service → Books Appointment or Gets Token → Receives Queue Position → Waiting Time Estimated → Staff Calls User → Service Completed → Analytics Updated

The project should not become only a basic appointment calendar.

It should combine:

- Appointments
- Walk-in digital queue
- Token generation
- Live queue position
- Waiting-time estimation
- Counter management
- Staff management
- Check-in
- No-show handling
- Notifications
- Dashboard and analytics
- Web/mobile deployment

Additional AI Features

Teams may also implement any of the following features to make their project more complete and demonstrate stronger technical implementation during evaluation:

Waiting-Time Prediction Predict waiting time using historical service duration and live queue conditions.	Peak-Hour Prediction Identify the busiest hours or days for each department.
No-Show Prediction Estimate which appointments may be more likely to be missed based on previous patterns.	Smart Staff Recommendation Suggest how many counters or staff members should be active based on expected demand.
Service Demand Forecasting Predict which services may receive higher demand on specific days.	AI Queue Optimization Recommend how customers can be distributed between counters to reduce overall waiting time.
AI Management Insights Generate short insights such as: "Document Verification experiences its longest queues between 11 AM and 1 PM."	

These additional AI features are not mandatory, but they can help teams demonstrate a more intelligent and complete solution during judging.

12. Submission Method

Each team must submit the following:

1. Application Demonstration Video

Submit a short video showing the complete working application.

The video should demonstrate:

- Appointment booking
- Walk-in token generation
- Live queue
- Waiting-time estimation
- Counter/staff workflow
- Token calling
- Check-in and completion
- Dashboard and analytics

2. GitHub Link OR Deployed Application Link

Teams must submit at least one of the following:

Option A — GitHub Repository Link

Submit the complete project source code.

OR

Option B — Deployed Application Link

Submit a working public link where the application can be tested.

For a mobile application, teams may submit an APK file or installation/testing link.

Teams may submit both if they want.

3. Project Explanation Document

Submit a separate PDF or Word document explaining the structure of the project.

The document should briefly explain:

- Important folders
- Important files
- Purpose of each folder/file
- Where appointment and queue logic is implemented
- Where any AI or intelligent feature is implemented
- How the frontend, backend, database, and other main parts connect

The document does not need to explain every line of code. It should simply help the judges understand how the project is organized.